

# PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

Anonymous ACL submission

## Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. Public Text2Cypher benchmarks rarely reflect private schemas, terminology, access patterns, or difficulty distributions. We present PIPE-Cypher, a local-model pipeline for generating organization-specific natural-language-to-Cypher benchmarks. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained Cypher generation, deterministic read-only and schema validation, execution feedback, repair, diversity controls, and LLM-judge review. The system is designed for industry settings where benchmark generation must avoid paid APIs, preserve data governance, and remain repeatable as graphs evolve.

## 1 Introduction

Property graphs encode enterprise relationships directly: account transfers, identity entitlements, access paths, customer interactions, and fraud rings. Cypher is a common query language for these graphs. As LLMs become natural-language interfaces for graph analytics, organizations need reliable benchmarks for natural-language-to-Cypher systems on their own schemas.

Static public benchmarks are insufficient for this setting. They cannot contain private labels, relationship types, categorical values, or domain-specific wording, and they do not track schema evolution. PIPE-Cypher addresses this gap by generating private, repeatable, execution-grounded NL-Cypher benchmarks.

## 2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), and Text2Cypher (Ozsoy et al., 2025), show that

high-quality Cypher data requires more than asking an LLM for query pairs. These works motivate schema grounding, execution checks, synthetic generation, verification, and complexity-aware evaluation. PIPE-Cypher differs by targeting private enterprise benchmark generation as the primary artifact: organizations bring their own graph, run local models, enforce Cypher constraints, and receive an executable benchmark with diversity and judge metadata.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) motivate realistic database tasks, execution-based evaluation, and enterprise workflow complexity. CIKM AutoQuery (Zheng et al., 2024) motivates strategy-level workload generation analysis and separating workload quality from model quality. LDBC FinBench (Qi et al., 2023) and SNB (Püroja et al., 2023) provide graph workloads suitable for industry-oriented evaluation.

For generated benchmark quality, PIPE-Cypher combines text-generation diversity diagnostics with text-to-query structure metrics. Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) capture lexical repetition, while schema coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells capture properties that matter for executable Cypher benchmarks.

## 3 Method

PIPE-Cypher has five stages: schema profiling, template generation, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. Deterministic checks enforce read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. A

lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

## 4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Local models are served through a vLLM/OpenAI-compatible endpoint on ds-serv6, using Qwen3.5 models for generation and judging.

The intended full-quality model is Qwen3.5-35B-A3B, but our June 1, 2026 live evidence uses the cached Qwen3.5-9B fallback. The 35B-A3B weights were staged on ds-serv6 under root-backed storage. A serving-capacity check estimated that the staged weights require four A5000 GPUs under our vLLM budget, while only one GPU was safely free in the live snapshot, so the reported generation and downstream results use the 9B fallback.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher uses seeded workload templates with deterministic reverse-binding queries and a mixed mode that prepends proven workload seeds to LLM-proposed templates. Every run records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name par-

tials, and small typos.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B fallback-model prompts within context limits on larger schemas.

The deterministic Cypher layer borrows from production lessons in the BalkanID Cypher system: schema-only prompting, exact matching, relationship direction discipline, `RETURN DISTINCT`, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as `UNION`, `CALL`, `UNWIND`, `WHERE EXISTS`, multiple `WHERE` clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

## 5 Experiments

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB. Categories include simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k.

## 6 Results

The full live run produced 3,000 accepted examples from 4,777 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted full-run records were exported into a benchmark package with stable example identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash for artifact tracking. The export contains 3,000 accepted examples: 2,000 FinBench, 1,000 SNB, and

| Gate                              | Evidence checked                                           | Failure mode caught                       |
|-----------------------------------|------------------------------------------------------------|-------------------------------------------|
| Read-only safety                  | blocked Cypher write/admin tokens                          | destructive or operational queries        |
| Schema validity                   | labels, relationship types, properties, categorical values | hallucinated graph vocabulary or values   |
| Direction validity                | observed relationship directions                           | reversed or invalid traversals            |
| Cypher analysis and normalization | structure extraction, rewrite safety, RETURN DISTINCT      | unsafe rewrites or duplicate/noisy rows   |
| Question constraints              | quoted values use exact literals                           | fuzzy matching of exact user values       |
| Execution                         | query runs and returns rows                                | syntactic validity without answerability  |
| LLM judge                         | ambiguity, semantic alignment, schema use                  | executable but semantically weak examples |

Table 1: PIPE-Cypher candidate acceptance gates.

| Graph    | Target/category | Binding limit | Min seed capacity | All categories meet target |
|----------|-----------------|---------------|-------------------|----------------------------|
| FinBench | 250             | 300           | 300               | Yes                        |
| SNB      | 125             | 200           | 200               | Yes                        |

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

| Setting                | Value                               |
|------------------------|-------------------------------------|
| Accepted examples      | 3,000                               |
| Primary graph          | LDBC FinBench, 2,000 examples       |
| Secondary graph        | LDBC SNB, 1,000 examples            |
| Categories             | 8 balanced categories, 375 each     |
| Generation/judge model | Local Qwen3.5-9B fallback           |
| Execution backend      | Neo4j Community, two live databases |
| Judge audit packet     | 80 sampled accepted/rejected pairs  |

Table 3: Full live experimental setup used for the generated benchmark artifact.

375 accepted examples in every planned category.

We report diversity as a diagnostic, not only as a design target. The full export has perfect category balance and near-perfect difficulty balance, uses 1,115 unique grounded entity values, and exactly quotes grounded values in 82.6% of examples with entity bindings. The Qwen3.5-9B fallback run still has low query-signature diversity because seeded graph-grounded templates are doing substantial work. This is useful evidence for industry deployment: the artifact is balanced and executable, while the appendix makes template and value concentration visible rather than hiding it.

The full-run failure taxonomy is concentrated in diversity/duplicate controls during recovery and empty execution results; only 2/4,777 candidates were schema-invalid after deterministic Cypher governance.

For judge calibration, the released artifacts include an 80-row audit packet sampled from accepted and rejected full-run candidates, with a 40/40 judge accept/reject split and coverage over

| Graph    | Candidates | Accepted | Acceptance | Categories at target |
|----------|------------|----------|------------|----------------------|
| FinBench | 3,404      | 2,000    | 0.588      | 8/8                  |
| SNB      | 1,373      | 1,000    | 0.728      | 8/8                  |
| Total    | 4,777      | 3,000    | 0.628      | 16/16                |

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

| Export artifact     | Examples | FinBench | SNB   | Train | Dev | Test |
|---------------------|----------|----------|-------|-------|-----|------|
| Live full benchmark | 3,000    | 2,000    | 1,000 | 2,408 | 296 | 296  |

Table 5: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash 8bc79a53a06b291a.

both graphs and all eight categories. The calibration tooling tracks graph, category, difficulty, strategy, and judge-decision coverage, and reports agreement, precision/recall, specificity, balanced accuracy, and Cohen’s kappa once human labels are filled. The current draft treats these labels as pending human review rather than as a generation gate.

Finally, we ran a downstream Text2Cypher evaluation using local Qwen3.5-9B on the exported full test split. The model saw schema text and the natural-language question, generated Cypher, and was evaluated by live execution against the corresponding FinBench or SNB database. The result verifies that the exported artifact can drive downstream model evaluation and gives a difficulty signal: the zero-shot 9B baseline solves simple aggregation examples but struggles on more operational categories.

## 7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The generated artifact records the schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for ev-

| Artifact property                   | Value                     |
|-------------------------------------|---------------------------|
| Categories                          | 8 balanced categories     |
| FinBench/category                   | 250                       |
| SNB/category                        | 125                       |
| Difficulty split                    | 1,521 easy / 1,479 medium |
| Unique labels / rel. types          | 7 / 9                     |
| Read/syntax/schema/exec/judge gates | 3,000/3,000               |

Table 6: Distribution and gate summary for the exported full benchmark artifact.

| Split          | Examples | Parse | Schema | Exec. success | Exec. acc. | Answer F1 |
|----------------|----------|-------|--------|---------------|------------|-----------|
| Live full test | 296      | 0.959 | 0.905  | 0.622         | 0.189      | 0.189     |

Table 7: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

ery example. Long-running jobs can be monitored from JSONL records and recovered with category-specific top-up runs that reject questions accepted in earlier runs. This design supports private benchmark refreshes without exposing schemas or values to paid APIs, while still leaving audit hooks for data owners to sample accepted and rejected examples.

## 8 Conclusion

PIPE-Cypher provides a practical path for enterprises to create private, executable, balanced NL-to-Cypher benchmarks. The pipeline combines schema introspection, constrained generation, deterministic validation, execution feedback, diversity control, and automated judge review.

## Limitations

Execution validity does not guarantee semantic correctness. LLM judges can over-accept plausible but wrong examples, so PIPE-Cypher releases a frozen post-hoc audit packet and requires complete human labels before reporting judge-human agreement. Generated artifacts may contain private schema details or sampled values, so organizations should apply normal data governance controls.

## Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

## References

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| Metric                                  | Value                 |
|-----------------------------------------|-----------------------|
| Question Distinct-1                     | 0.041                 |
| Question Distinct-2                     | 0.088                 |
| Question self-BLEU-2 (sampled)          | 0.826                 |
| Unique query-signature ratio            | 0.010                 |
| Top query-signature share               | 0.083                 |
| Category normalized entropy             | 1.000                 |
| Graph-category normalized entropy       | 0.980                 |
| Difficulty normalized entropy           | 1.000                 |
| Label coverage                          | 0.412                 |
| Relationship-type coverage              | 0.375                 |
| Property-name coverage                  | 0.407                 |
| Unique grounded-value ratio             | 0.373                 |
| Grounded values exactly quoted          | 0.826                 |
| Aggregation / negation / ordering rates | 0.500 / 0.125 / 0.125 |

Table 8: Diversity diagnostics for the full exported benchmark. Distinct-n follows text-generation usage; self-BLEU is lower when questions are less redundant; grounded-value metrics are aggregate-only and do not list raw values.

| Failure bucket              | Count | Share of rejected |
|-----------------------------|-------|-------------------|
| Diversity/duplicate control | 1,335 | 0.751             |
| Empty result                | 374   | 0.210             |
| Judge semantic reject       | 66    | 0.037             |
| Schema invalid              | 2     | 0.001             |

Table 9: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

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| Setting           | Graph    | Records | Accepted | Acceptance | Categories at target |
|-------------------|----------|---------|----------|------------|----------------------|
| Unconstrained LLM | FinBench | 0       | 0        | 0.000      | 0/8                  |
| Unconstrained LLM | SNB      | 0       | 0        | 0.000      | 0/8                  |
| Reverse-only      | FinBench | 400     | 400      | 1.000      | 8/8                  |
| Reverse-only      | SNB      | 402     | 400      | 0.995      | 8/8                  |
| Validators+repair | FinBench | 400     | 400      | 1.000      | 8/8                  |
| Validators+repair | SNB      | 402     | 400      | 0.995      | 8/8                  |
| No retrieval      | FinBench | 429     | 400      | 0.932      | 8/8                  |
| No retrieval      | SNB      | 402     | 400      | 0.995      | 8/8                  |
| No rewrite        | FinBench | 423     | 400      | 0.946      | 8/8                  |
| No rewrite        | SNB      | 402     | 400      | 0.995      | 8/8                  |
| No LLM judge      | FinBench | 401     | 400      | 0.998      | 8/8                  |
| No LLM judge      | SNB      | 406     | 400      | 0.985      | 8/8                  |
| Full PIPE-Cypher  | FinBench | 416     | 400      | 0.962      | 8/8                  |
| Full PIPE-Cypher  | SNB      | 402     | 400      | 0.995      | 8/8                  |

Table 10: Live target-50 ablation evidence with local Qwen3.5-9B. Each graph run targets 50 accepted examples per category.

| Setting           | Graph    | Read-only | Syntax | Schema | Exec. | Judge |
|-------------------|----------|-----------|--------|--------|-------|-------|
| Unconstrained LLM | FinBench | 0.000     | 0.000  | 0.000  | 0.000 | 0.000 |
| Unconstrained LLM | SNB      | 0.000     | 0.000  | 0.000  | 0.000 | 0.000 |
| Reverse-only      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 1.000 |
| Reverse-only      | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| Validators+repair | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 1.000 |
| Validators+repair | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| No retrieval      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.932 |
| No retrieval      | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| No rewrite        | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.946 |
| No rewrite        | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| No LLM judge      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.998 |
| No LLM judge      | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.985 |
| Full PIPE-Cypher  | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.962 |
| Full PIPE-Cypher  | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |

Table 11: Quality-gate rates for the live target-50 ablation suite. Rates are computed over all generated records in each graph/setting.

## A Additional Results

## B Claim–Evidence Map

This section maps the main paper claims to the strongest current evidence and to the remaining risks. This is intended as a reviewer-facing audit surface: claims with pending evidence remain marked as such rather than being folded into the main results.

- Claim 1.** PIPE-Cypher can generate a private, executable NL-to-Cypher benchmark over enterprise-style property graphs using local models.

*Evidence.* The full Qwen3.5-9B fallback run exported 3,000 accepted examples: 2,000 FinBench and 1,000 SNB, with 375 examples in every planned workload category and all accepted examples passing read-only, syntax, schema, execution, non-empty-result, and judge gates.

*Key artifacts.* benchmark export; manifest snapshot; paper table: benchmark export; paper table: full results

*Status.* Supported by live 9B fallback evidence; 35B target run is blocked by current GPU capacity.

| Audit packet property | Value     |
|-----------------------|-----------|
| Rows                  | 80        |
| Judge accept / reject | 40 / 40   |
| FinBench / SNB rows   | 48 / 32   |
| Easy / medium rows    | 26 / 54   |
| Labeled rows          | 0         |
| Calibration status    | unlabeled |

Table 12: Post-hoc judge calibration packet coverage. Human labels are pending until the packet is complete and are not used as a generation gate.

| Metric             | Point | 95% CI         | SE    | N   |
|--------------------|-------|----------------|-------|-----|
| Parse valid        | 0.959 | [0.936, 0.980] | 0.012 | 296 |
| Schema valid       | 0.905 | [0.872, 0.939] | 0.017 | 296 |
| Execution success  | 0.622 | [0.564, 0.676] | 0.028 | 296 |
| Execution accuracy | 0.189 | [0.145, 0.233] | 0.022 | 296 |
| Answer F1          | 0.189 | [0.149, 0.236] | 0.023 | 296 |

Table 13: Bootstrap uncertainty for downstream Text2Cypher evaluation on the full exported test split. Intervals use row-level resampling with a fixed seed and are intended for appendix reporting.

*Risk.* Generation quality may change with the target 35B model or with private enterprise schemas beyond FinBench/SNB.

- Claim 2.** Cypher-specific governance makes generated benchmark candidates safer and more auditable than prompt-only generation.

*Evidence.* The pipeline validates read-only safety, schema-visible labels, node and relationship properties, relationships, observed relationship direction, categorical values, contextual return columns, parser-style structure, execution success, and LLM-judge review. In the full run, only 2 of 4,777 candidates were schema-invalid after deterministic governance, and all 3,000 exported examples passed the deterministic and judge gates.

*Key artifacts.* code: validator.py; code: cypher\_parser.py; code: question\_constraints.py; snapshot: failure taxonomy; +1 more

*Status.* Supported by code, tests, and full-run failure taxonomy.

*Risk.* Semantic correctness still depends on execution evidence and judge review; human audit labels are pending.

| Downstream outcome | Count | Share of incorrect |
|--------------------|-------|--------------------|
| Answer mismatch    | 128   | 0.533              |
| Execution failed   | 72    | 0.300              |
| Schema invalid     | 28    | 0.117              |
| Parse invalid      | 12    | 0.050              |

Table 14: Downstream Text2Cypher failure taxonomy for local Qwen3.5-9B on the full exported test split. Shares exclude exact-answer matches.

- Claim 3.** Reverse grounding and structured workload seeds are designed to improve reliability under local-model constraints.

*Evidence.* The target-50 FinBench/SNB ablation suite completed 14/14 graph/variant cells, reached every category target for all non-empty/non-unconstrained runs, and passed the paper-readiness audit with logs, model IDs, code revision, run summaries, and gate-rate availability. Appendix-only tables plus yield and gate-rate figures report the audited target-50 evidence while the target-100 follow-up and seeded target-50 repeat continue for stronger final reliability and variance claims.

*Key artifacts.* script: run live ablation suite; script: monitor remote ablation suite; script: monitor remote ablation queue; script: collect remote ablation suite; +9 more

*Status.* Supported by audited target-50 appendix evidence; target-100 and seeded target-50 repeat runs are still pending for stronger final reliability claims.

*Risk.* A single target-50 suite is the minimum publishable ablation scale, not the strongest possible evidence. If target-100 or repeated-seed runs complete, final claims should use those sensitivity results instead of relying only on target-50.

- Claim 4.** The benchmark controls category and difficulty balance while exposing residual diversity limits.

*Evidence.* The full export has perfect category balance, planned 2:1 FinBench/SNB graph mix, and a near-even easy/medium split. Diversity diagnostics report Distinct-n, sampled self-BLEU-2, query-signature diversity, normalized entropy, schema coverage, structural feature rates, and aggregate value-grounding diagnostics. The full export uses 1,115 unique



|     |                                                         |                                                       |     |
|-----|---------------------------------------------------------|-------------------------------------------------------|-----|
| 427 | grounded entity values and exactly quotes               | rejects, both graphs, and all eight categories.       | 475 |
| 428 | grounded values in 82.6% of examples with               | Deterministic sidecar generation now creates          | 476 |
| 429 | entity bindings, making template and value              | two independent annotator sheets plus an ad-          | 477 |
| 430 | concentration visible instead of hidden.                | judication template, with a tracked hash mani-        | 478 |
| 431 | <i>Key artifacts.</i> snapshot: diversity report; paper | fest that does not commit raw value-bearing           | 479 |
| 432 | table: diversity; paper figure: diversity diag-         | sheets. The analyzer reports agreement, preci-        | 480 |
| 433 | nostics; paper figure: full export distribution         | sion/recall, specificity, balanced accuracy, and      | 481 |
| 434 | <i>Status.</i> Supported by full-export statistics and  | Cohen’s kappa only after the packet is com-           | 482 |
| 435 | diversity diagnostics.                                  | pletely human-labeled; partial labels are not         | 483 |
| 436 | <i>Risk.</i> Low query-signature diversity in the 9B    | treated as paper-ready calibration.                   | 484 |
| 437 | fallback run remains a limitation and ablation          | <i>Key artifacts.</i> code: judge.py; code: calibra-  | 485 |
| 438 | target.                                                 | tion.py; code: judge_audit_packet.py; script:         | 486 |
|     |                                                         | create judge annotation sheets; +6 more               | 487 |
| 439 | 5. <b>Claim 5.</b> The generated benchmark is useful    | <i>Status.</i> Pipeline gate is implemented; audit    | 488 |
| 440 | for downstream Text2Cypher evaluation.                  | packet coverage is supported; judge-human             | 489 |
| 441 | <i>Evidence.</i> A zero-shot local Qwen3.5-9B           | agreement remains unreported until all audit          | 490 |
| 442 | downstream evaluation over the 296-example              | rows are labeled.                                     | 491 |
| 443 | full test split reports parse validity, schema va-      | <i>Risk.</i> Judge-human agreement is not yet         | 492 |
| 444 | lidity, execution success, execution accuracy,          | known because human labels are still blank.           | 493 |
| 445 | answer F1, per-category breakdowns, fixed-              |                                                       |     |
| 446 | seed bootstrap uncertainty intervals, and a             | 7. <b>Claim 7.</b> The intended 35B local genera-     | 494 |
| 447 | row-level error taxonomy. The model reaches             | tion/judge path is staged but not currently           | 495 |
| 448 | 0.189 execution accuracy with a 95% boot-               | runnable under observed shared-GPU con-               | 496 |
| 449 | strap interval of [0.145, 0.233] and 0.622 ex-          | straints.                                             | 497 |
| 450 | ecution success with interval [0.564, 0.676].           | <i>Evidence.</i> Qwen3.5-35B-A3B is staged on ds-     | 498 |
| 451 | The error taxonomy classifies 240 incorrect             | serv6, but the capacity checker reports 68,573        | 499 |
| 452 | rows into answer mismatches, execution fail-            | MiB of weights, four required A5000 GPUs              | 500 |
| 453 | ures, schema invalid predictions, and parse-            | under the conservative vLLM budget, and               | 501 |
| 454 | invalid predictions, showing the benchmark              | only one safe GPU in the latest snapshot.             | 502 |
| 455 | can discriminate easy aggregation from harder           | <i>Key artifacts.</i> script: check vllm capacity;    | 503 |
| 456 | operational categories and can explain model            | script: render vllm capacity snapshot; re-            | 504 |
| 457 | failure modes.                                          | search note: model availability; research note:       | 505 |
| 458 | <i>Key artifacts.</i> downstream summary; snap-         | qwen35b capacity snapshot 20260601; +1                | 506 |
| 459 | shot: downstream uncertainty; snapshot:                 | more                                                  | 507 |
| 460 | downstream error report; research note: down-           | <i>Status.</i> Documented blocker; current paper      | 508 |
| 461 | stream evaluation; +6 more                              | results are explicitly reported as Qwen3.5-9B         | 509 |
| 462 | <i>Status.</i> Supported by live downstream evalua-     | fallback evidence.                                    | 510 |
| 463 | tion against FinBench/SNB databases, with               | <i>Risk.</i> A later 35B run may change yield, diver- | 511 |
| 464 | bootstrap uncertainty computed from row-                | sity, judge behavior, and downstream conclu-          | 512 |
| 465 | level outputs.                                          | sions.                                                | 513 |
| 466 | <i>Risk.</i> Only one downstream model has been         |                                                       |     |
| 467 | evaluated so far.                                       | <b>C Prompt Contracts</b>                             | 514 |
| 468 | 6. <b>Claim 6.</b> Automated LLM-judge review can       | PIPE-Cypher treats prompts as versioned imple-        | 515 |
| 469 | replace human-in-the-loop generation gates              | mentation artifacts. The list summarizes the prompt   | 516 |
| 470 | while still exposing judge risk.                        | contracts used for generation, repair, judging, and   | 517 |
| 471 | <i>Evidence.</i> The pipeline uses deterministic val-   | downstream evaluation; hashes fingerprint the full    | 518 |
| 472 | idation plus LLM-judge review as the gener-             | prompt constants in the codebase.                     | 519 |
| 473 | ation gate. A post-hoc 80-row judge audit               | 1. <b>Template generation.</b> Stage: Workload pro-   | 520 |
| 474 | packet preserves 40 judge accepts, 40 judge             | posal. SHA-256: 10b25b.                               | 521 |

|     |                                                      |                                                                                                                                                                                                                                        |     |
|-----|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 522 | <i>Contract.</i> Schema-only labels, relationships,  | natural-language question, accepted Cypher, struc-                                                                                                                                                                                     | 566 |
| 523 | properties, and categorical values; Realistic        | tural tags, gate status, and a bounded execution-                                                                                                                                                                                      | 567 |
| 524 | enterprise analyst wording; At most two typed        | result sample.                                                                                                                                                                                                                         | 568 |
| 525 | slots and JSON-only output                           |                                                                                                                                                                                                                                        |     |
| 526 | 2. <b>Reverse binding.</b> Stage: Graph grounding.   | 1. <b>FinBench / Boolean Existence / easy. Ques-</b>                                                                                                                                                                                   | 569 |
| 527 | SHA-256: 48e949.                                     | <i>tion:</i> Does account '187743809466009406'                                                                                                                                                                                         | 570 |
| 528 | <i>Contract.</i>                                     | have any outgoing transfer?                                                                                                                                                                                                            | 571 |
| 529 | Read-only                                            |                                                                                                                                                                                                                                        |     |
| 530 | MATCH/WHERE/RETURN DIS-                              | <pre>MATCH (src:Account {accountId: '187743809466009406'}) -[:TRANSFER_TO]-&gt; (:Account) RETURN DISTINCT COUNT(DISTINCT src) &gt; 0 AS HasOutgoingTransfer</pre>                                                                     | 572 |
| 531 | TINCT/LIMIT only; Slot variables named               |                                                                                                                                                                                                                                        | 573 |
| 532 | exactly as requested; Forward relationship           |                                                                                                                                                                                                                                        | 574 |
|     | directions from the schema                           |                                                                                                                                                                                                                                        | 575 |
| 533 | 3. <b>Cypher generation.</b> Stage: Candidate query. | <i>Structure:</i> single_hop, aggregation.                                                                                                                                                                                             | 579 |
| 534 | SHA-256: 61c557.                                     | <i>Relationships:</i> TRANSFER_TO.                                                                                                                                                                                                     | 580 |
| 535 | <i>Contract.</i> Only schema-visible constructs and  | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                                | 581 |
| 536 | observed directions; RETURN DISTINCT for             | <i>Result sample:</i> {HasOutgoingTransfer: True};                                                                                                                                                                                     | 582 |
| 537 | set returns and exact equality for quoted val-       | observed rows: 1                                                                                                                                                                                                                       | 583 |
| 538 | ues; Context columns, categorical hints, place-      |                                                                                                                                                                                                                                        |     |
| 539 | holderized retrieval, and no writes                  | 2. <b>FinBench / Complex Aggregation / medium.</b>                                                                                                                                                                                     | 584 |
| 540 | 4. <b>Repair.</b> Stage: Validation feedback. SHA-   | <i>Question:</i> What is the total transferred amount                                                                                                                                                                                  | 585 |
| 541 | 256: 56c419.                                         | from accounts owned by person 'Gwar'?                                                                                                                                                                                                  | 586 |
| 542 | <i>Contract.</i> Preserve question intent while fix- | <pre>MATCH (p:Person {personName: 'Gwar'}) -[:OWN_ACCOUNT]-&gt; (src:Account)-[:TRANSFER_TO]-&gt; (:Account) RETURN DISTINCT SUM(t.amount) AS TotalTransferredAmount</pre>                                                             | 587 |
| 543 | ing validation or execution issues; Keep query       |                                                                                                                                                                                                                                        | 588 |
| 544 | read-only and schema-grounded; Return only           | <i>Structure:</i> join_heavy, aggregation.                                                                                                                                                                                             | 594 |
| 545 | corrected Cypher                                     | <i>Relationships:</i> OWN_ACCOUNT, TRANS-                                                                                                                                                                                              | 595 |
| 546 | 5. <b>LLM judge.</b> Stage: Quality gate. SHA-256:   | FER_TO.                                                                                                                                                                                                                                | 596 |
| 547 | 421c7b.                                              | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                                | 597 |
| 548 | <i>Contract.</i> Inputs include question, Cypher,    | <i>Result sample:</i> {TotalTransferredAmount:                                                                                                                                                                                         | 598 |
| 549 | schema slice, execution rows, and validation         | 14972318.41}; observed rows: 1                                                                                                                                                                                                         | 599 |
| 550 | summary; Strict JSON scores for ambiguity,           |                                                                                                                                                                                                                                        |     |
| 551 | semantic alignment, schema use, and diffi-           | 3. <b>FinBench / Complex Retrieval / easy. Ques-</b>                                                                                                                                                                                   | 600 |
| 552 | culty; Categorical values constrain query liter-     | <i>tion:</i> Which accounts received transfers from                                                                                                                                                                                    | 601 |
| 553 | als, not observed result-row values; Pass only       | accounts owned by person 'Zof'?                                                                                                                                                                                                        | 602 |
| 554 | useful, unambiguous enterprise benchmark             | <pre>MATCH (p:Person {personName: 'Zof'}) -[:OWN_ACCOUNT]-&gt; (src:Account) -[:TRANSFER_TO]-&gt; (dst:Account) RETURN DISTINCT dst.accountId AS AccountId, dst.accountType AS AccountType, dst.isBlocked AS IsBlocked LIMIT 300</pre> | 603 |
| 555 | examples                                             |                                                                                                                                                                                                                                        | 604 |
| 556 | 6. <b>Downstream Text2Cypher.</b> Stage: Model       | <i>Structure:</i> join_heavy, bounded_result.                                                                                                                                                                                          | 612 |
| 557 | evaluation. SHA-256: 4c07ff.                         | <i>Relationships:</i> OWN_ACCOUNT, TRANS-                                                                                                                                                                                              | 613 |
| 558 | <i>Contract.</i> Read-only Cypher only; Schema-      | FER_TO.                                                                                                                                                                                                                                | 614 |
| 559 | visible constructs and exact direction preser-       | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                                | 615 |
| 560 | vation; RETURN DISTINCT, count/ranking               | <i>Result sample:</i> {AccountId:                                                                                                                                                                                                      | 616 |
| 561 | rules, and no explanations                           | 4687402787162554854, AccountType:                                                                                                                                                                                                      | 617 |
| 562 |                                                      | credit card, IsBlocked: False}; observed rows:                                                                                                                                                                                         | 618 |
| 563 |                                                      | 3                                                                                                                                                                                                                                      | 619 |
| 564 |                                                      |                                                                                                                                                                                                                                        |     |
| 565 |                                                      |                                                                                                                                                                                                                                        |     |
| 562 | <b>D Representative Accepted Examples</b>            |                                                                                                                                                                                                                                        |     |
| 563 | The examples below are selected in stable identifier |                                                                                                                                                                                                                                        |     |
| 564 | order from the tracked full-export snapshot, one per |                                                                                                                                                                                                                                        |     |
| 565 | graph/category cell when available. They show the    |                                                                                                                                                                                                                                        |     |



#### 4. FinBench / Negation Difference / medium.

*Question:* Which accounts owned by person 'Kant' have not sent any transfers?

```
MATCH (p:Person {personName:
'Kant'})
-[:OWN_ACCOUNT]-> (a:Account)
WHERE NOT (a) -[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT a.accountId AS
AccountId, a.accountType AS
AccountType,
a.isBlocked AS IsBlocked
LIMIT 300
```

*Structure:* join\_heavy, negation, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4739757132830738341, AccountType: merchant account, IsBlocked: False}; observed rows: 1

#### 5. FinBench / Path Temporal / medium. *Question:*

Which accounts can receive money within two transfer hops from accounts owned by person 'Sossamon'?

```
MATCH (p:Person {personName:
'Sossamon'}) -[:OWN_ACCOUNT]->
(src:Account) -[:TRANSFER_TO*1..2]->
(dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS
IsBlocked
LIMIT 300
```

*Structure:* join\_heavy, path, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False}; observed rows: 4

#### 6. FinBench / Ranking Topk / medium. *Question:*

For accounts owned by person 'Barry', which account sent the highest total transfer amount?

```
MATCH (p:Person {personName:
'Barry'})
-[:OWN_ACCOUNT]->
(src:Account) -[:TRANSFER_TO]->
(:Account)
WITH src, SUM(t.amount) AS
totalAmount
RETURN DISTINCT src.accountId AS
AccountId, src.accountType AS
AccountType, src.isBlocked AS
IsBlocked,
totalAmount
```

```
ORDER BY totalAmount DESC
LIMIT 1
```

*Structure:* join\_heavy, aggregation, order\_rank, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4732438783436260772, AccountType: internet account, IsBlocked: False}; observed rows: 1

#### 7. FinBench / Simple Aggregation / easy. *Question:*

How many accounts are owned by person 'Kaewsuktae'?

```
MATCH (p:Person {personName:
'Kaewsuktae'}) -[:OWN_ACCOUNT]->
(a:Account)
RETURN DISTINCT COUNT(DISTINCT a) AS
AccountCount
```

*Structure:* single\_hop, aggregation.

*Relationships:* OWN\_ACCOUNT.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountCount: 1}; observed rows: 1

#### 8. FinBench / Simple Retrieval / easy. *Question:*

Which accounts are owned by person 'Barry'?

```
MATCH (p:Person {personName:
'Barry'})
-[:OWN_ACCOUNT]-> (a:Account)
RETURN DISTINCT a.accountId AS
AccountId, a.accountType AS
AccountType,
a.isBlocked AS IsBlocked
LIMIT 300
```

*Structure:* single\_hop, bounded\_result.

*Relationships:* OWN\_ACCOUNT.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4732438783436260772, AccountType: internet account, IsBlocked: False}; observed rows: 1

#### 9. SNB / Boolean Existence / medium. *Question:*

Does person with id 6597069766828 like any post?

```
MATCH (p:Person {id:
6597069766828})
OPTIONAL MATCH (p) -[:LIKES]->
(post:Post)
RETURN DISTINCT COUNT(post) > 0 AS
LikesAnyPost
```

|     |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 730 | <i>Structure:</i> single_hop, aggregation, optional.                                                                                                                                                                              | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                                                                        | 783                                                                       |
| 731 | <i>Relationships:</i> LIKES.                                                                                                                                                                                                      | <i>Result sample:</i> {FirstName: K., LastName: Sen, PersonId: 94}; observed rows: 1                                                                                                                                                                                           | 784<br>785                                                                |
| 732 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                |                                                                           |
| 733 | <i>Result sample:</i> {LikesAnyPost: True}; observed rows: 1                                                                                                                                                                      | 13. <b>SNB / Path Temporal / easy. Question:</b> Which people are within two knows hops of person with id 4398046511136?                                                                                                                                                       | 786<br>787<br>788                                                         |
| 734 |                                                                                                                                                                                                                                   | <pre>MATCH (src:Person {id: 4398046511136}) -[:KNOWS*1..2]-&gt; (dst:Person) RETURN DISTINCT dst.id AS PersonId, dst.firstName AS FirstName, dst.lastName AS LastName AS LastName LIMIT 200</pre>                                                                              | 789<br>790<br>791<br>792<br>793<br>794<br>795<br>796                      |
| 735 | 10. <b>SNB / Complex Aggregation / medium. Question:</b> How many distinct posts are in forums joined by person with id 6597069766845?                                                                                            | <i>Structure:</i> single_hop, path, bounded_result.                                                                                                                                                                                                                            | 797                                                                       |
| 736 | <pre>MATCH (forum:Forum) -[:HAS_MEMBER]-&gt; (p:Person {id: 6597069766845}) MATCH (forum) -[:CONTAINER_OF]-&gt; (post:Post) RETURN DISTINCT COUNT(DISTINCT post) AS JoinedForumPostCount</pre>                                    | <i>Relationships:</i> KNOWS.                                                                                                                                                                                                                                                   | 798                                                                       |
| 737 |                                                                                                                                                                                                                                   | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                                                                        | 799                                                                       |
| 738 | <i>Structure:</i> join_heavy, aggregation.                                                                                                                                                                                        | <i>Result sample:</i> {FirstName: Rafael, LastName: Fernández, PersonId: 4398046511333}; observed rows: 25                                                                                                                                                                     | 800<br>801<br>802                                                         |
| 739 | <i>Relationships:</i> CONTAINER_OF, HAS_MEMBER.                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 740 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                           | 14. <b>SNB / Ranking Topk / medium. Question:</b> Among forums tagged 'James_K._Polk', which forum has the most members?                                                                                                                                                       | 803<br>804<br>805                                                         |
| 741 | <i>Result sample:</i> {JoinedForumPostCount: 1}; observed rows: 1                                                                                                                                                                 | <pre>MATCH (forum:Forum) -[:HAS_TAG]-&gt; (tag:Tag {name: 'James_K._Polk'}) MATCH (forum) -[:HAS_MEMBER]-&gt; (person:Person) WITH forum, COUNT(DISTINCT person) AS memberCount RETURN DISTINCT forum.title AS ForumTitle, memberCount ORDER BY memberCount DESC LIMIT 1</pre> | 806<br>807<br>808<br>809<br>810<br>811<br>812<br>813<br>814<br>815<br>816 |
| 742 |                                                                                                                                                                                                                                   | <i>Structure:</i> join_heavy, aggregation, order_rank, bounded_result.                                                                                                                                                                                                         | 817<br>818                                                                |
| 743 |                                                                                                                                                                                                                                   | <i>Relationships:</i> HAS_MEMBER, HAS_TAG.                                                                                                                                                                                                                                     | 819                                                                       |
| 744 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                           | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                                                                        | 820                                                                       |
| 745 | <i>Result sample:</i> {PersonId: 4398046511220}; observed rows: 19                                                                                                                                                                | <i>Result sample:</i> {ForumTitle: Wall of Javed Khan, memberCount: 33}; observed rows: 1                                                                                                                                                                                      | 821<br>822                                                                |
| 746 |                                                                                                                                                                                                                                   | 15. <b>SNB / Simple Aggregation / easy. Question:</b> How many posts are tagged 'Vietnam'?                                                                                                                                                                                     | 823<br>824                                                                |
| 747 | 11. <b>SNB / Complex Retrieval / easy. Question:</b> Which people are members of forums containing posts tagged 'Manuel_Noriega'?                                                                                                 | <pre>MATCH (post:Post) -[:HAS_TAG]-&gt; (tag:Tag {name: 'Vietnam'}) RETURN DISTINCT COUNT(DISTINCT post) AS PostCount</pre>                                                                                                                                                    | 825<br>826<br>827<br>828<br>829<br>830                                    |
| 748 | <pre>MATCH (forum:Forum) -[:HAS_MEMBER]-&gt; (p:Person), (forum) -[:CONTAINER_OF]-&gt; (post:Post) -[:HAS_TAG]-&gt; (tag:Tag {name: 'Manuel_Noriega'}) RETURN DISTINCT p.id AS PersonId LIMIT 200</pre>                           | <i>Structure:</i> single_hop, aggregation.                                                                                                                                                                                                                                     | 831                                                                       |
| 749 | <i>Structure:</i> join_heavy, bounded_result.                                                                                                                                                                                     | <i>Relationships:</i> HAS_TAG.                                                                                                                                                                                                                                                 | 832                                                                       |
| 750 | <i>Relationships:</i> CONTAINER_OF, HAS_MEMBER, HAS_TAG.                                                                                                                                                                          | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                                                                        | 833                                                                       |
| 751 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                                                                                                                                                                                           | <i>Result sample:</i> {PostCount: 1}; observed rows: 1                                                                                                                                                                                                                         | 834<br>835                                                                |
| 752 | <i>Result sample:</i> {PersonId: 4398046511220}; observed rows: 19                                                                                                                                                                |                                                                                                                                                                                                                                                                                |                                                                           |
| 753 | 12. <b>SNB / Negation Difference / medium. Question:</b> Which members of forum 'Wall of Celso Oliveira' have not liked any post?                                                                                                 |                                                                                                                                                                                                                                                                                |                                                                           |
| 754 | <pre>MATCH (forum:Forum {title: 'Wall of Celso Oliveira'}) -[:HAS_MEMBER]-&gt; (p:Person) WHERE NOT (p) -[:LIKES]-&gt; (:Post) RETURN DISTINCT p.id AS PersonId, p.firstName AS FirstName, p.lastName AS LastName LIMIT 200</pre> |                                                                                                                                                                                                                                                                                |                                                                           |
| 755 | <i>Structure:</i> join_heavy, negation, bounded_result.                                                                                                                                                                           |                                                                                                                                                                                                                                                                                |                                                                           |
| 756 | <i>Relationships:</i> HAS_MEMBER, LIKES.                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                |                                                                           |
| 757 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 758 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 759 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 760 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 761 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 762 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 763 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 764 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 765 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 766 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 767 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 768 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 769 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 770 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 771 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 772 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 773 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 774 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 775 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 776 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 777 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 778 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 779 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 780 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 781 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |
| 782 |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                                                                           |

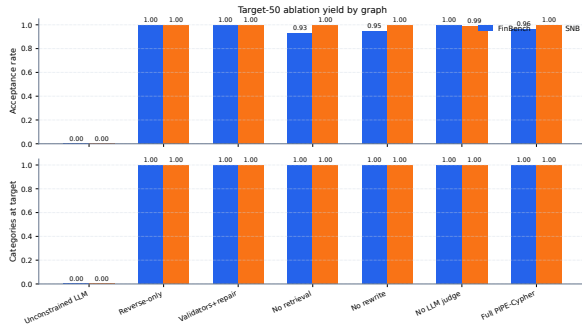


Figure 1: Audited target-50 ablation suite over FinBench and SNB. The suite reaches all graph/variant/category targets and is included as appendix evidence while the larger target-100 and seeded repeat suites continue running for stronger final reliability evidence.

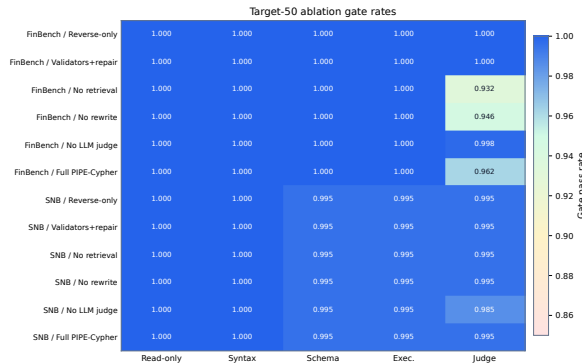


Figure 2: Gate-rate heatmap for the audited target-50 ablation suite. Deterministic read-only and syntax gates are saturated, while schema, execution, and judge rates expose the remaining quality differences across graph and pipeline variants.

16. **SNB / Simple Retrieval / easy.** *Question:* Which post IDs did person with id 4398046511124 like?

```
MATCH (p:Person {id:
4398046511124})
-[:LIKES]-> (post:Post)
RETURN DISTINCT post.id AS PostId
LIMIT 200
```

*Structure:* single\_hop, bounded\_result.

*Relationships:* LIKES.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {PostId: 343597385744}; observed rows: 5

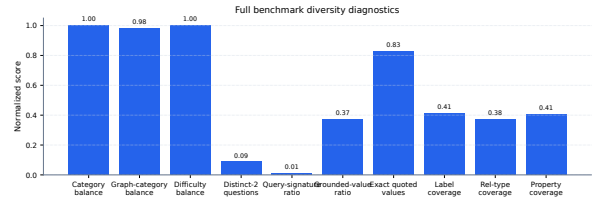


Figure 3: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the fallback seeded-template run.

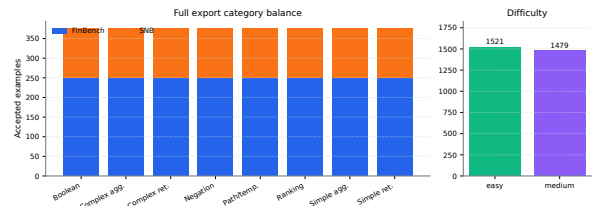


Figure 4: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

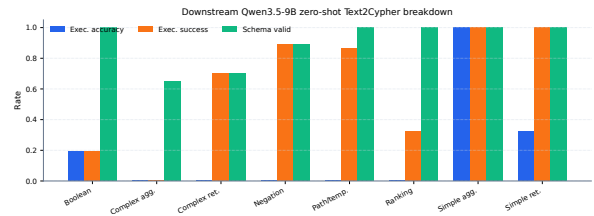


Figure 5: Downstream zero-shot Qwen3.5-9B Text2Cypher evaluation by workload category on the full test split. The breakdown separates final execution accuracy from parse/schema/execution success signals, making difficult operational categories visible.

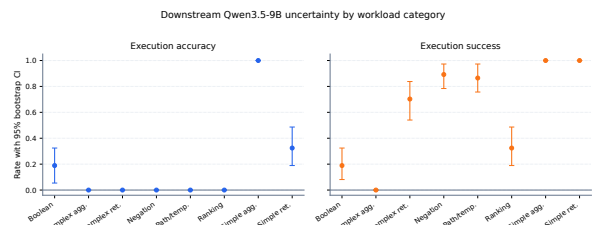


Figure 6: Bootstrap uncertainty for downstream Qwen3.5-9B Text2Cypher evaluation by workload category. Error bars show 95% row-level bootstrap intervals over the full exported test split.

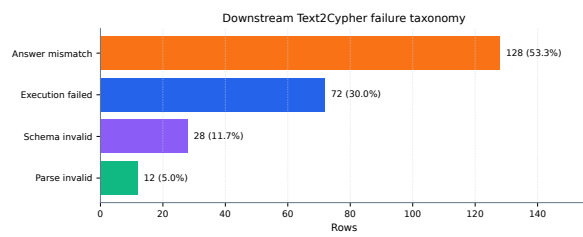


Figure 7: Downstream Text2Cypher failure taxonomy for the local Qwen3.5-9B baseline on the full test split. Exact matches are excluded so the chart exposes whether misses come from invalid Cypher, failed execution, or semantically wrong executable queries.